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Data Science and Data Visualization

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FINAL REPORT

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VISUALIZING GLOBAL HAPPINESS TREND

(2010 - 2023)

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I. Introduction

1.1 Context

In an era of globalization and shifting geopolitical landscapes, measuring a nation's success solely by economic output (GDP) is increasingly viewed as insufficient. The World Happiness Report provides a more holistic metric known as the "Life Ladder"—a subjective measure of well-being where citizens rank their current lives on a scale of 0 to 10. Understanding the drivers of this score is crucial for policymakers, economists, and social scientists.

1.2 Dataset

This project utilizes data from the World Happiness Report, covering the period from 2010 to 2023. The dataset includes approximately 150 countries and tracks several key metrics:

- **Life Ladder:** The primary dependent variable (Happiness score).
- **Log GDP per Capita:** A measure of economic output.
- **Social Support:** The perception of having someone to count on in times of trouble.
- **Healthy Life Expectancy:** Physical health metrics.
- **Freedom to Make Life Choices:** Perception of autonomy.
- **Generosity and Perceptions of Corruption.**

1.3 Project Objective

The objective of this project was to engineer a web-based, interactive data visualization dashboard using D3.js. Unlike static reports, this tool allows users to explore the temporal evolution of happiness, perform multi-country comparisons, and identify correlations between economic factors and well-being dynamically.

II. Research Questions

To guide the design of the visualization, we established four core research questions that the dashboard must answer:

Geospatial Clustering (The "Neighbor Effect"):

- **Question:** Is the "Life Ladder" score spatially correlated? Do high-happiness countries cluster together geographically, or is happiness randomly distributed?

Regional Influence:

- **Question:** To what extent does the macro-region (e.g., Europe, Africa, Asia) determine the distribution of happiness scores? Is there a distinct "ceiling" or "floor" for certain regions?

Correlations and Drivers:

- **Question:** How do specific attributes, specifically Log GDP per Capita and Social Support, correlate with the Life Ladder? Does a higher GDP guarantee a higher Life Ladder score?

Temporal Trends:

- **Question:** Which regions or specific nations have shown significant improvement or decline over the last decade (2010–2023)? Can we identify volatility in specific countries?

III. Data Processing & Pipeline

Reliable visualization requires reliable data. The raw dataset contained gaps, inconsistencies in naming conventions, and sparse records for certain nations. We implemented a robust data processing pipeline using Python (Pandas) to prepare the data for the D3.js frontend.

3.1 Data Cleaning & Filtering

The raw data was filtered to include only the years 2010 through 2023.

- **Constraint:** To ensure the time-series analysis (Line Chart) was legible and accurate, we enforced a rule to exclude countries with fewer than 7 recorded years of data.
- **Rationale:** Countries with sporadic data points (e.g., only 2 or 3 years of records) create misleading trend lines and contribute to visual "noise" without adding statistical value.

3.2 Region Mapping & Standardization

The raw dataset provided specific sub-regions (e.g., "Western Europe", "Sub-Saharan Africa"). To facilitate higher-level aggregation in the Stacked Bar Chart, we mapped all countries into 5 Macro-Regions:

- **Africa**
- **The Americas (North & South)**
- **Asia (including the Middle East)**
- **Europe**
- **Oceania**

GeoJSON alignment: Country names in the CSV were programmatically renamed (e.g., "Congo, Dem. Rep." to "Democratic Republic of the Congo") to match the topological keys in the GeoJSON file, ensuring 100% match rates for the map rendering.

3.3 Data Imputation (Linear Interpolation)

A critical challenge in the dataset was missing years. For example, a country might have data for 2012 and 2014, but not 2013.

Technique: We applied Linear Interpolation to fill these gaps.

Rationale:

- **Visual Stability:** Without interpolation, the D3 time-slider animation would cause countries to "flicker" (disappear and reappear) on the map.
- **Trend Continuity:** In the line charts, gaps result in broken lines. Linear interpolation provides a reasonable estimate for the missing intermediate years, creating smooth, continuous time-series lines that are easier for the human eye to track.

3.4 Categorical Binning

To support the Stacked Bar Chart, we converted the continuous "Life Ladder" variable into discrete categories:

- **Low:** Life Ladder < 5.0
- **Medium:** Life Ladder between 5.0 and 7.0
- **High:** Life Ladder > 7.0

This allows users to quickly assess the quality of life in a region rather than just the average score.

IV. Visualization Design & Techniques

To effectively communicate the multi-dimensional nature of the dataset (geospatial, temporal, and categorical), we implemented a coordinated multiple-view (CMV) dashboard comprising three distinct but interconnected visualizations.

4.1 The Choropleth Map (Geospatial View)

The primary entry point for exploration is a world map.

- **Technique:** We utilized D3.js geoPath with a Mercator projection to render the geometry. To ensure accessibility for users with color vision deficiencies (CVD), we employed the Viridis color scale (d3.interpolateViridis) for continuous variables like "Life Ladder" and "Log GDP.".
- **Rationale:** Spatial patterns in happiness are often contiguous. The map allows users to instantly identify regional clusters, such as the high-happiness block in Scandinavia versus the low-happiness belt in Sub-Saharan Africa, that would be lost in a standard table or scatter plot.

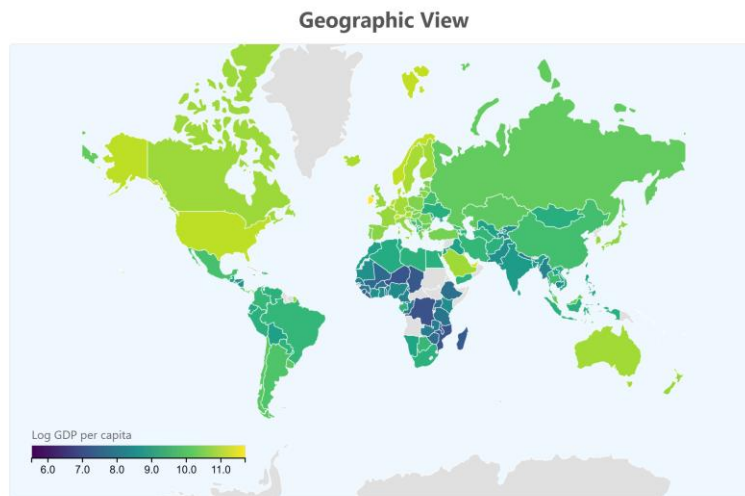


Figure 4.1.1. The choropleth graph with Viridis color scale

4.2 The Stacked Bar Chart (Regional Distribution)

While the map shows where happiness is distributed, the bar chart explains how it is distributed within macro-regions.

- **Technique:** A stacked bar chart groups countries by their region (x-axis) and stacks them based on the categorical "Life Ladder" bins (Low, Medium, High).
- **Color Strategy:** We adopted the Okabe-Ito palette (Vermilion, Yellow, Sky Blue), a color-blind-safe scheme that provides clear contrast between categories without relying on red-green distinction.
- **Rationale:** This chart solves the problem of "aggregation bias." A simple average score for "Asia" might hide the fact that it contains both extremely happy and extremely unhappy nations. The stacked layout preserves this granularity.

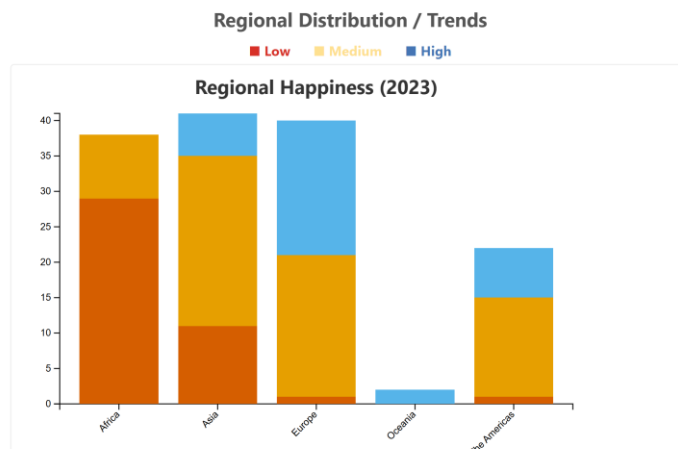


Figure 4.2.1. The stacked bar chart

4.3 The Multi-Series Line Chart (Temporal Trends)

To analyze change over time, we engineered a dynamic line chart.

Technique: This chart supports multi-series plotting. When a user selects a country, the system plots three lines simultaneously:

- The Country Trend (Solid Blue)
- The Regional Average (Dashed Orange)
- The World Average (Dotted Gray)

Rationale: A standalone trend line lacks context. By superimposing regional and global benchmarks, the visualization allows users to determine if a country's decline is an isolated incident or part of a global recession.

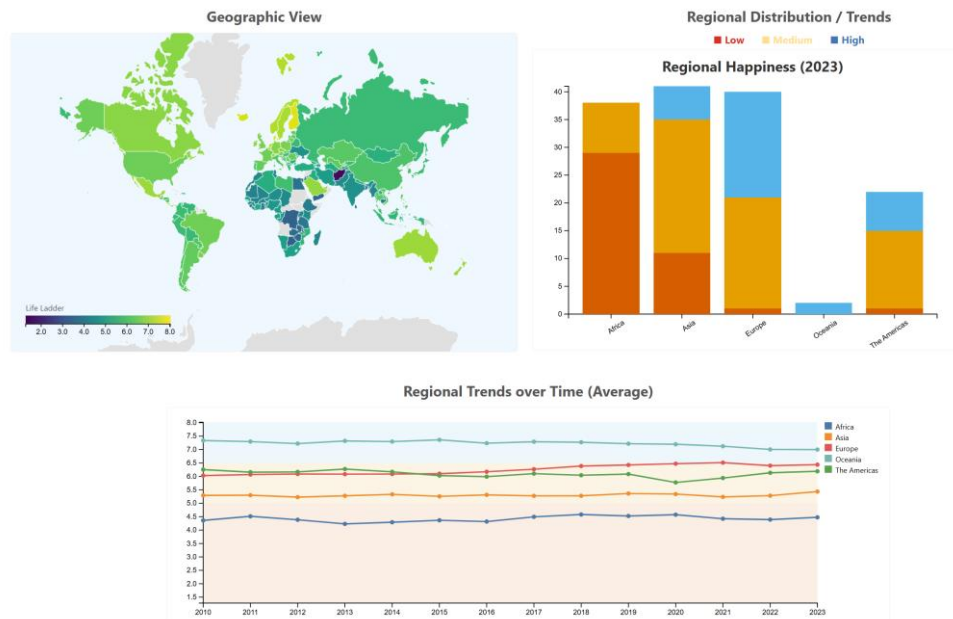


Figure 4.3.1. The multi-series line chart

V. Interaction Techniques & Benefits

Static charts are limited by the available screen real estate. To overcome this, we implemented "Details-on-Demand" through advanced interaction techniques.

5.1 Semantic Zooming (Drill-Down Logic)

The dashboard follows Ben Shneiderman's mantra: "Overview first, zoom and filter, then details-on-demand."

- **Macro View:** The default state shows global averages and regional stacks.
- **Meso View (Drill-Down):** Clicking a bar segment (e.g., "High Happiness in Europe") filters the data to display only the specific countries in that category, effectively zooming the user's cognitive focus.
- **Micro View:** Clicking a specific country reveals the granular year-by-year time series.

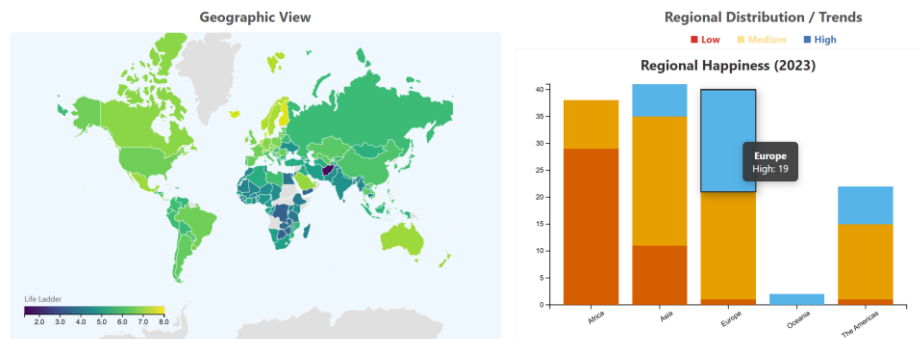


Figure 5.1.1. Macro view

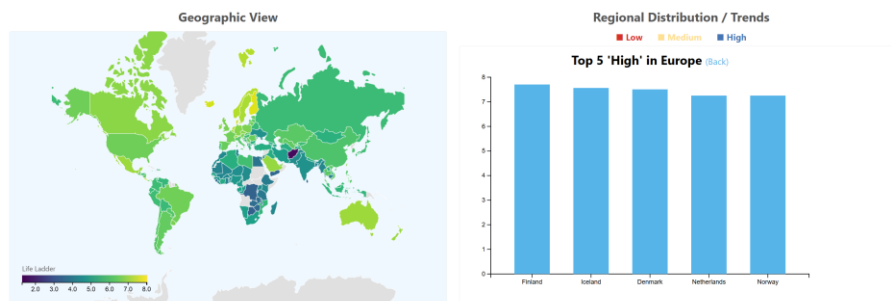


Figure 5.1.2. Meso view

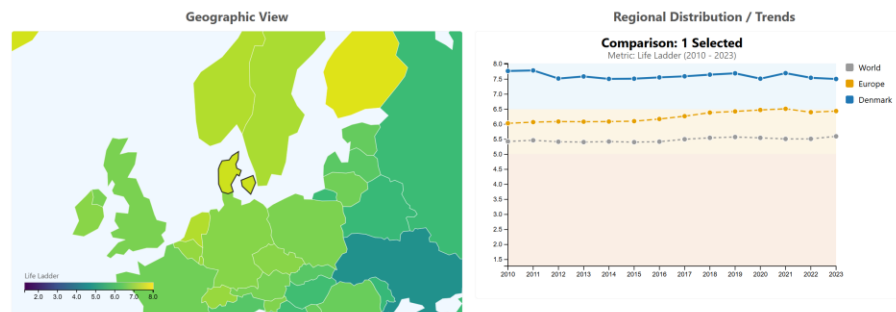


Figure 5.1.3. Micro view

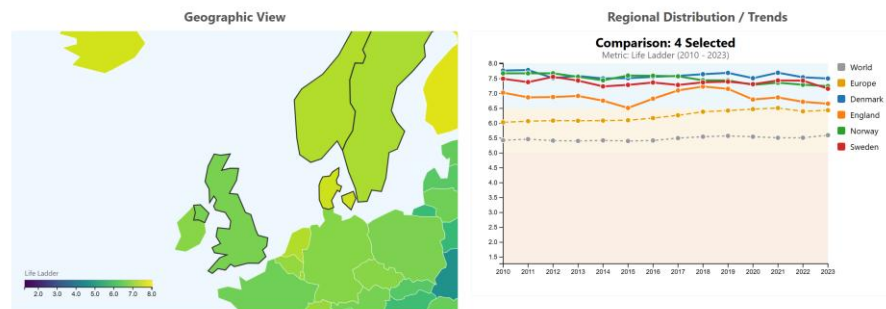


Figure 5.1.4. Micro view with other countries

5.2 Linked Selection & Brushing

All views are coupled. Selecting "Brazil" on the map instantly highlights South America in the regional chart and plots Brazil's specific economic trajectory in the line chart. This cohesion reduces cognitive load, as users do not need to manually cross-reference different charts.

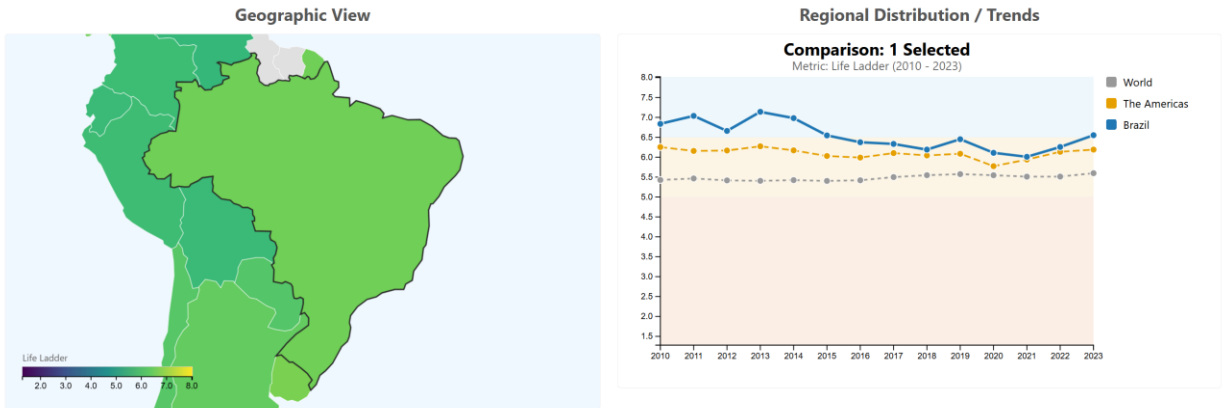


Figure 5.2.1. Linked selection

5.3 Geometric Zooming & Panning

Standard maps often obscure small nations. We implemented d3.zoom behaviors allowing users to pan and zoom into specific territories. To maintain visual clarity, stroke widths are dynamically re-scaled during zooming, preventing borders from becoming overwhelmingly thick at high magnification.

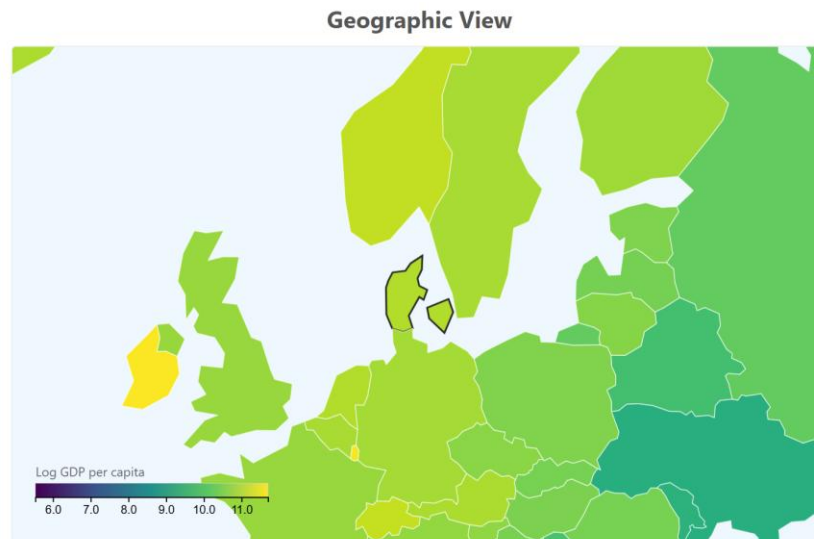


Figure 5.1.1. Zoom in

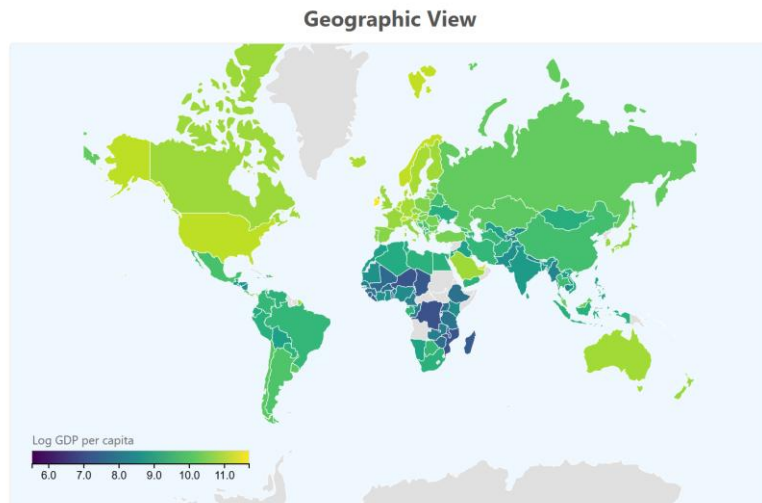


Figure 5.1.2. Zoom out

VI. Findings & Analysis

Using the dashboard, we derived several key insights regarding the state of global happiness:

6.1 Spatial Autocorrelation

There is strong evidence of the "Neighbor Effect." High "Life Ladder" scores are not randomly distributed; they form distinct clusters in Western Europe, North America, and Oceania. Conversely, regions experiencing political instability often form contiguous low-happiness zones, suggesting that well-being is heavily influenced by regional geopolitical stability.

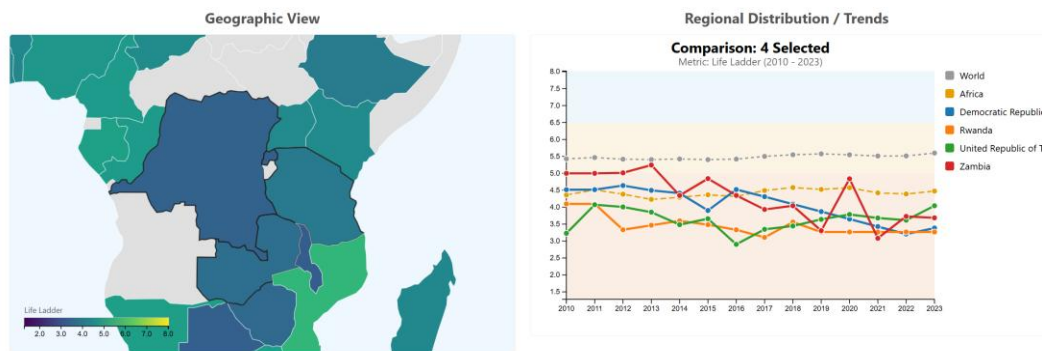


Figure 6.1.1. A cluster in Africa

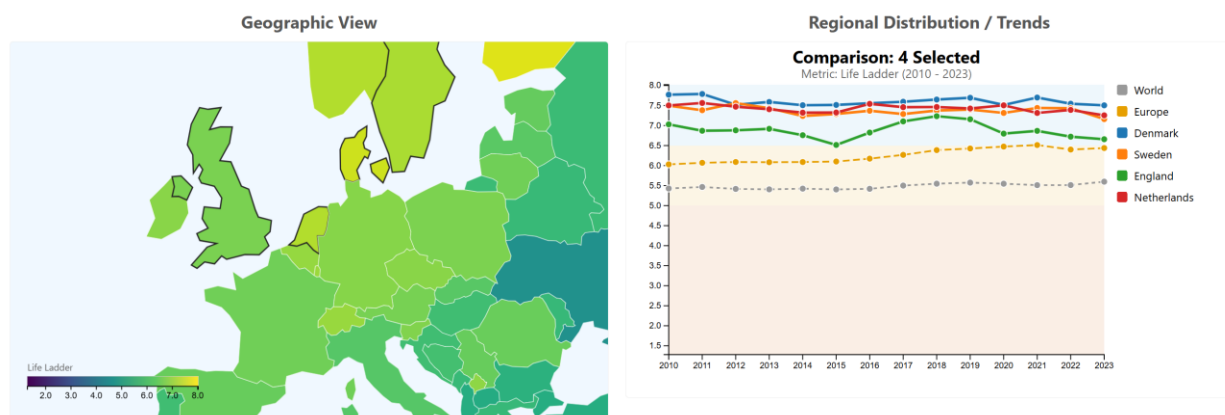


Figure 6.1.2. A cluster in Europe

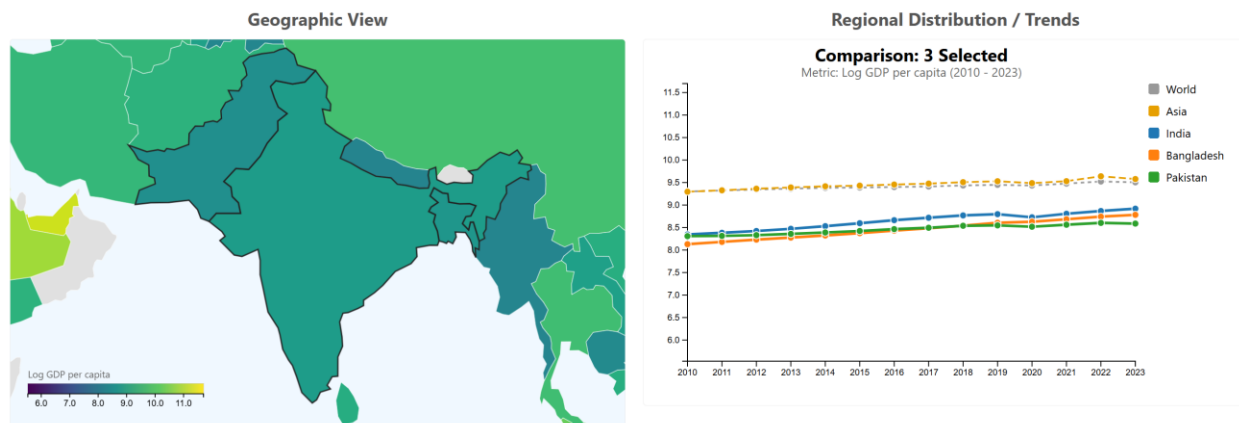


Figure 6.1.3. A cluster in Asia

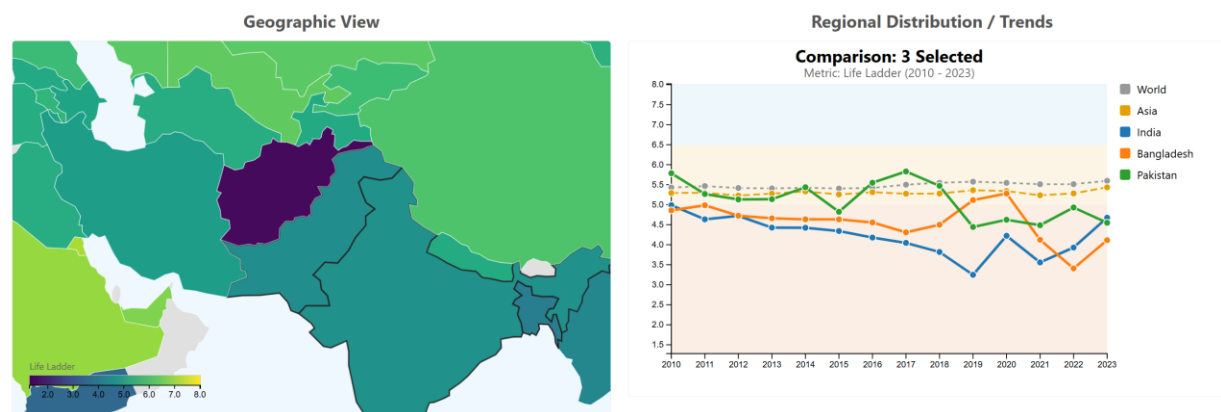


Figure 6.1.4. A same pattern also in GDP

6.2 The "Happiness Ceiling"

The Regional Bar Chart reveals a striking disparity: Europe and The Americas have almost zero representation in the "Low Happiness" category. In contrast, Africa is dominated by the "Low" category, with very few outliers breaking into the "High" tier. This indicates that geographic location remains a statistically significant predictor of life satisfaction.

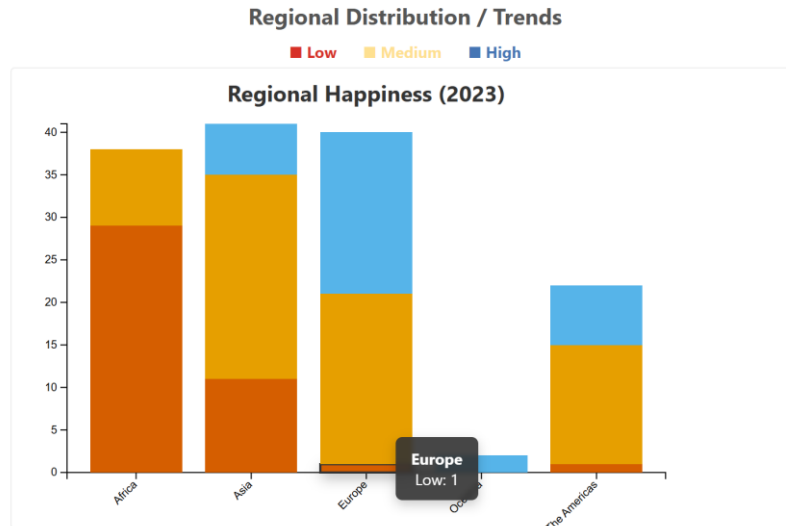


Figure 6.1.2. The regional bar chart

Interpretation of results: In 2023, Haiti (The Americans) experienced escalating gang violence, political vacuum, and mass displacement, generating acute humanitarian needs despite emerging international aid initiatives. Meanwhile, Ukraine's (Europe) summer counteroffensive yielded limited territorial gains amid heavy casualties, as continued Russian assaults on infrastructure intensified civilian hardship.

6.3 Economic Correlation vs. Anomaly

When switching the metric to Log GDP per Capita, the map closely mirrors the "Life Ladder" map, confirming a strong positive correlation between wealth and happiness. However, anomalies exist. North American countries report GDP higher than their happiness scores would predict.

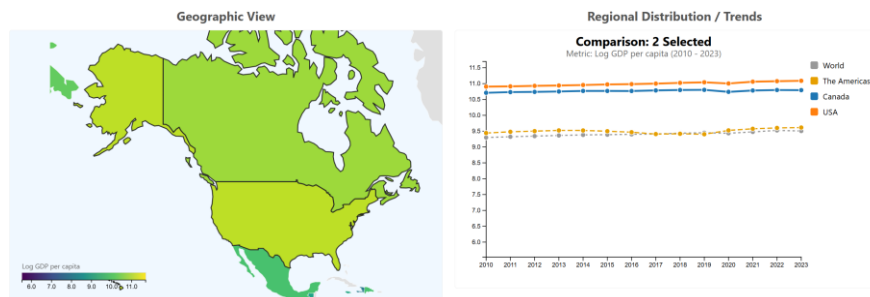


Figure 6.3.1. The GDP is not change in USA and Canada

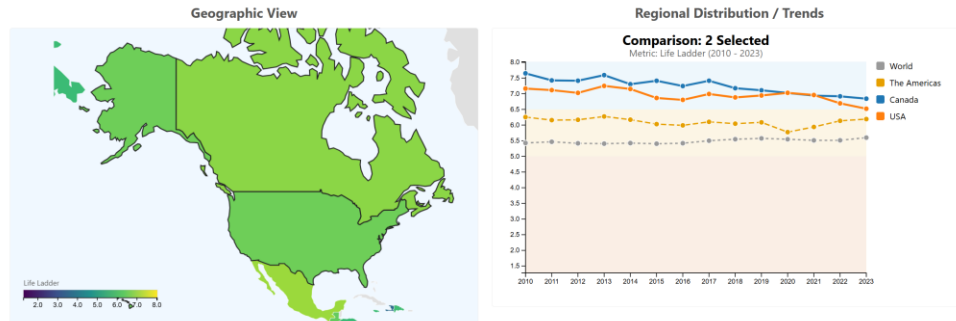


Figure 6.3.2. The Life Ladder is decrease in USA in Canada

6.4 Volatility in Conflict Zones

The Time Series view highlights extreme volatility in specific nations. For example, comparing Afghanistan against the "South Asia" average reveals a sharp divergence post-2017, visually quantifying the impact of political regime change on citizen well-being.

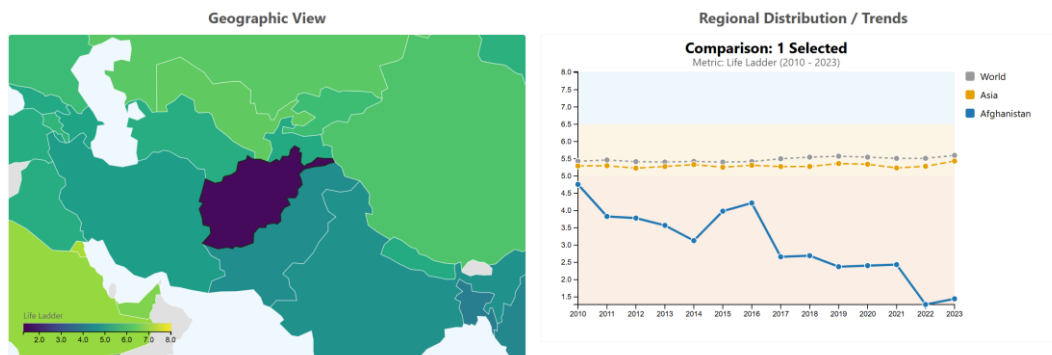


Figure 6.4.1. Afghanistan Life Ladder trend

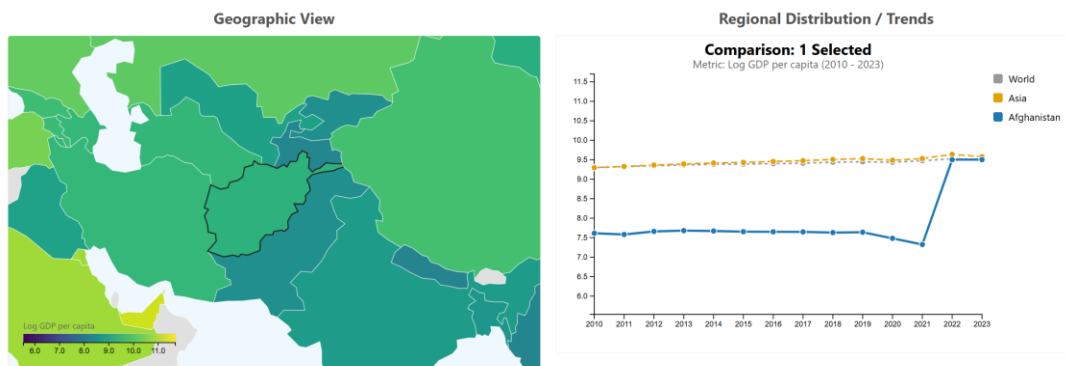


Figure 6.4.2. Afghanistan's GDP trend

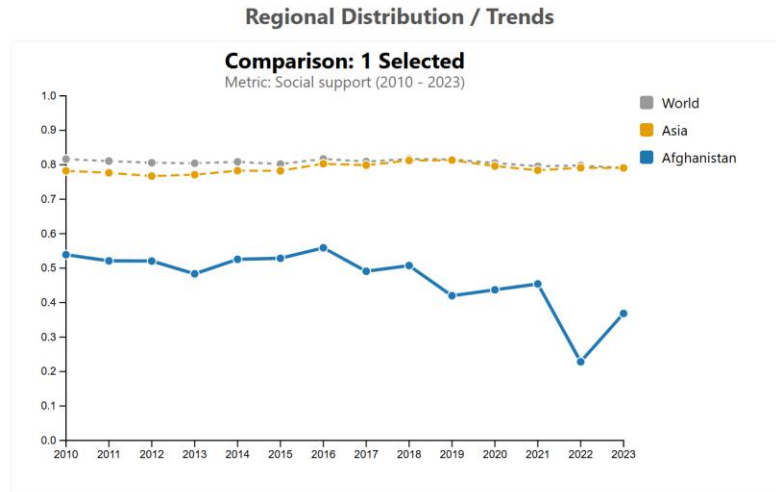


Figure 6.4.3. Afghanistan Social Support trend

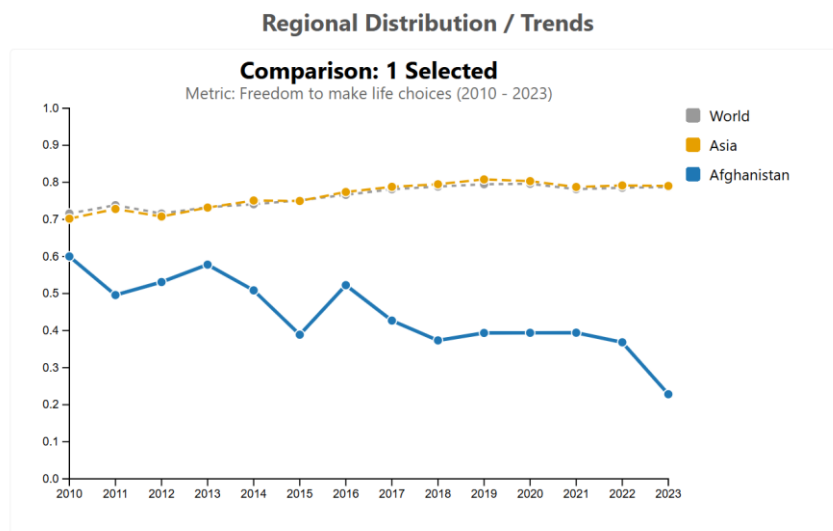


Figure 6.4.4. Afghanistan Freedom to make life choices trend

Interpretation of results: GDP does not fully capture variations in the Life Ladder. Declines in freedom to make life choices and in social support correspond with decreases in Life Ladder scores, even when Afghanistan experienced a surge in GDP growth during 2021–2022.

VII. Conclusion

This project successfully demonstrates that interactive data visualization can transform complex, multi-dimensional datasets into accessible narratives. By combining rigorous data engineering (imputation, cleaning) with user-centered design (accessible color scales, drill-down interactions), the dashboard provides a tool that is not just descriptive, but exploratory. It allows users to move beyond simple rankings and understand the nuanced, interconnected drivers of human happiness.

Reference

Dataset: World Happiness Data 2024 | Emotions Analysis

Code: DuNhan1930/DSDV_Project

Reference: DuNhan1930/Data-mining-world-happiness-2024

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